ROLE: Cloud / DevOps Engineer

OVERVIEW
Cloud / DevOps Engineers design, build, deploy, and maintain scalable cloud infrastructure and automated delivery pipelines. They bridge development and operations by ensuring applications are reliably deployed, highly available, secure, and cost-efficient. Their core goal is to enable fast, safe, and repeatable software delivery using cloud platforms like AWS, Azure, or GCP.

CORE RESPONSIBILITIES
- Design and implement cloud infrastructure (AWS/Azure/GCP)
- Build Infrastructure-as-Code (IaC) using Terraform or CloudFormation
- Develop and maintain CI/CD pipelines for automated deployments
- Deploy and manage applications in production environments
- Ensure system scalability, high availability, and fault tolerance
- Monitor infrastructure health, logs, and application performance
- Implement security best practices across cloud environments
- Manage backups, disaster recovery, and system reliability
- Automate infrastructure provisioning and configuration
- Collaborate with development, security, and operations teams
- Optimize cloud costs through right-sizing and usage analysis

REQUIRED PROGRAMMING & SYSTEM SKILLS
Cloud engineers must understand both software and systems deeply. Linux is mandatory because most cloud servers run Linux-based environments. Networking fundamentals are critical for understanding VPCs, routing, and security groups. Python or Bash is used for automation scripts. Git is essential for version control and CI/CD workflows.
- Linux administration and shell scripting
- Basic Python or Bash scripting for automation
- Networking fundamentals (DNS, TCP/IP, HTTP/HTTPS)
- Git and version control workflows
- System design fundamentals for scalable architecture

CLOUD PLATFORMS
AWS is the most widely used platform in industry, followed by Azure in enterprise environments and GCP in data/AI-heavy systems. Understanding at least one platform deeply is required; AWS is usually preferred for jobs.
- AWS (EC2, S3, RDS, Lambda, VPC, IAM, EKS)
- Microsoft Azure (VMs, App Services, Azure DevOps, AKS)
- Google Cloud Platform (Compute Engine, GKE, Cloud Run)
- Multi-cloud and hybrid cloud architecture concepts

INFRASTRUCTURE AS CODE (IaC)
IaC is the foundation of modern cloud engineering. It allows infrastructure to be defined in code instead of manual configuration. Terraform is the industry standard for multi-cloud setups, while CloudFormation is AWS-native.
- Terraform for cloud-agnostic infrastructure provisioning
- AWS CloudFormation for AWS-native deployments
- Infrastructure versioning and reproducibility
- Modular infrastructure design

CI/CD AND DEVOPS PIPELINES
CI/CD pipelines automate build, test, and deployment processes. This reduces manual errors and speeds up software delivery. Tools include GitHub Actions, Jenkins, AWS CodePipeline, and GitLab CI.
- GitHub Actions / Jenkins for CI/CD automation
- AWS CodePipeline, CodeBuild, CodeDeploy
- Automated testing and deployment workflows
- Blue-green and rolling deployments

CONTAINERIZATION AND ORCHESTRATION
Containers package applications with all dependencies, ensuring consistency across environments. Kubernetes manages and scales these containers

automatically.
- Docker for application containerization
- Amazon ECR for container image storage
- Kubernetes (EKS) for orchestration and scaling
- Helm for Kubernetes application packaging
- Auto-scaling using HPA and cluster autoscaler

MONITORING AND OBSERVABILITY
Monitoring ensures systems are healthy and performant. Observability helps detect and debug issues in production environments.
- AWS CloudWatch for logs, metrics, and alarms
- AWS CloudTrail for audit logging
- Prometheus for metrics collection
- Grafana for dashboards and visualization
- ELK Stack (Elasticsearch, Logstash, Kibana) for log analysis
- Alerting and incident response systems

NETWORKING AND SECURITY
Security is a core responsibility in cloud environments. Engineers must design secure architectures and enforce access control.
- VPC design (subnets, route tables, gateways)
- Security Groups and Network ACLs
- IAM roles, policies, and least privilege access
- AWS WAF and AWS Shield for protection
- Encryption at rest and in transit
- Identity integration (Active Directory, SSO)

STORAGE AND DATABASES
Cloud systems rely heavily on scalable storage and managed databases.
- S3 for object storage and backups
- EBS for block storage
- RDS (MySQL, PostgreSQL, Aurora)
- DynamoDB for NoSQL workloads
- Backup and lifecycle management policies

AUTOMATION AND SERVER MANAGEMENT
Automation reduces manual effort and improves reliability. Server management includes provisioning, patching, and scaling systems.
- AWS Systems Manager (SSM) for automation and patching
- Auto Scaling Groups for dynamic scaling
- Configuration management tools (Ansible, if used)
- Infrastructure automation scripts

MLOPS / PLATFORM INTEGRATION (OPTIONAL BUT VALUABLE)
Modern cloud engineers often support ML and data workloads.
- Hosting ML models on AWS SageMaker
- Serving APIs using Lambda or ECS/EKS
- Managing data pipelines for ML systems
- Supporting RAG and LLM infrastructure

ENTRY LEVEL EXPECTATIONS
At entry level, cloud engineers are expected to deploy applications, manage basic AWS services, and understand CI/CD pipelines. Hands-on practice with EC2, S3, and IAM is essential.
- Basic AWS services (EC2, S3, IAM, VPC)
- Linux system basics
- Docker fundamentals
- Simple CI/CD pipeline setup
- Basic scripting (Python/Bash)
- Understanding of networking fundamentals

INTERMEDIATE LEVEL EXPECTATIONS
At intermediate level, engineers can design scalable architectures, manage Kubernetes clusters, and build production-grade CI/CD pipelines.

- Terraform or CloudFormation for IaC
- Kubernetes (EKS/AKS/GKE)
- Advanced CI/CD pipelines
- Monitoring and observability setup
- Secure cloud architecture design
- Multi-tier application deployment

ADVANCED LEVEL EXPECTATIONS
Advanced engineers design enterprise-scale systems, optimize cost and performance, and lead infrastructure decisions.
- High availability and fault-tolerant architectures
- Multi-region deployments
- Disaster recovery planning
- Cost optimization strategies (FinOps)
- Large-scale Kubernetes management
- Security compliance and governance

COMMON PROJECTS FOR PORTFOLIO
Projects that demonstrate real-world DevOps skills are highly valued.
- CI/CD pipeline for full-stack application deployment
- Kubernetes-based microservices deployment system
- Serverless application using AWS Lambda
- Scalable web app with auto-scaling infrastructure
- Monitoring dashboard with Prometheus + Grafana
- Infrastructure-as-Code project using Terraform
- Cloud-based RAG or AI deployment system

LEARNING ROADMAP FOR CLOUD / DEVOPS
Start with Linux and networking fundamentals. Learn AWS core services (EC2, S3, IAM, VPC). Move into Docker and CI/CD pipelines. Then learn Terraform and Kubernetes. Finally focus on monitoring, security, and system design. Build at least 3 real deployment projects.

RELATED ROLES
- DevOps Engineer
- Cloud Engineer
- Site Reliability Engineer (SRE)
- Platform Engineer
- Infrastructure Engineer
- Kubernetes Engineer
- Cloud Architect